

Data Quality: Principles & Industry Horror Stories



Suman

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Pre-Reads

Module-1

Recommended

Resource



Associated Lectures (1)



Description



Discussions

Session 4.3: Pre-read Material

Data Quality: Principles & Industry Horror Stories

Program: Vishlesan i-Hub IIT Patna x Masai School -- AIM (AI & Machine Learning) **Session ID:** 4.3 I

Week: 4 **Prerequisites:** Session 4.2 (Data Wrangling: Cleaning, Merging & GroupBy) **Session Duration:** 105 minutes **Preparation Time:** 20-30 minutes

What This Session Covers

This session shifts focus from the mechanics of data manipulation to the critical question every analyst and ML engineer must ask before any analysis begins: **"Is this data good enough to trust?"** The session covers seven topics:

The GIGO Principle -- "Garbage In, Garbage Out" and how machine learning amplifies data quality problems rather than compensating for them

The Six Data Quality Dimensions -- A systematic framework (accuracy, completeness, consistency, timeliness, validity, uniqueness) for evaluating any dataset, with Pandas code demonstrations for each

Industry Horror Stories -- Four real-world disasters caused by data quality failures: JPMorgan's \$6.2 billion Excel error, the UK Post Office wrongful convictions, Target's pregnancy prediction backlash, and gene name corruption in published research

Data Governance Basics -- Who owns data, who maintains quality, and how to track where data came from (ownership, stewardship, lineage, change management, access controls)

Documentation and Data Dictionaries -- How to describe a dataset so anyone can understand it without asking you

Regulatory Context -- Why data quality is a legal obligation (GDPR, HIPAA, SOX) and what the penalties look like

Connecting Data Quality to ML -- How everything you have learned in Weeks 3-4 maps directly to the machine learning pipeline, plus a hands-on mini-project building a Data Quality Audit Report

By the end of this session, you will be able to evaluate any dataset across six quality dimensions, detect common quality issues using Pandas, explain why data quality failures cause real-world disasters, and

produce a structured quality scorecard summarizing a dataset's fitness for analysis.

Prerequisites Checklist

Before attending this session, make sure you are comfortable with:

- ☐ Creating DataFrames using `pd.DataFrame()` with dictionaries (Session 4.1)
- ☐ Loading CSV files with `pd.read_csv()` and exploring data with `.head()`, `.info()`, `.describe()`, `.shape` (Session 4.1)
- ☐ Selecting columns with bracket notation and rows with `.loc[]` / `.iloc[]` (Session 4.1)
- ☐ Filtering rows with boolean conditions: `df[df['column'] > value]` (Session 4.1)
- ☐ Handling missing values with `.isnull()`, `.dropna()`, `.fillna()` (Session 4.2)
- ☐ Cleaning strings with `.str.lower()`, `.str.strip()`, `.str.replace()` (Session 4.2)
- ☐ Merging DataFrames with `pd.merge()` (Session 4.2)
- ☐ Aggregating data with `.groupby()` and aggregate functions like `.mean()`, `.sum()`, `.count()` (Session 4.2)
- ☐ Using `.value_counts()` and `.nunique()` to explore categorical columns (Session 4.2)
- ☐ All Python fundamentals and NumPy skills from Weeks 1-3

If any of these feel unclear, review your Session 4.1 and 4.2 materials before class.

Important note: Pandas is already installed from Session 4.1. No new packages are required for this session.

Business Story: IBM Watson for Oncology

Before you learn any frameworks or run any code, you need to understand what happens when data quality is ignored at the highest level -- with the best engineers, the biggest budgets, and the most prestigious partnerships in the world.

The Billion-Dollar AI That Gave Wrong Cancer Advice

In 2013, IBM launched one of the most ambitious AI projects in history: **Watson for Oncology** -- a system designed to recommend cancer treatments to doctors worldwide. IBM partnered with Memorial Sloan Kettering Cancer Center (MSK), one of the world's premier cancer hospitals, to train Watson. The vision was extraordinary: an AI that could bring world-class oncology expertise to every hospital on the planet.

The critical flaw was not in the algorithm. Watson was **not trained on real patient outcome data at scale**. Instead, it was trained on a small set of synthetic and hypothetical cancer cases -- essentially textbook examples curated by a handful of MSK physicians. The system learned what those specific doctors **THOUGHT** was the right treatment, not what actually worked across thousands of real patients with diverse backgrounds, co-morbidities, and genetic profiles.

The Consequences

In 2018, internal IBM documents revealed Watson had recommended **"unsafe and incorrect" treatment recommendations** in multiple cases, including suggesting treatments that could cause severe bleeding in patients already at risk. **MD Anderson Cancer Center** spent over \$62 million on a Watson partnership before terminating it. **Manipal Hospitals in India** found that Watson's recommendations often did not match clinical judgment for Indian patient populations, whose genetic profiles differ significantly from the Western patients in MSK's training data. Multiple hospitals across Southeast Asia, Europe, and South Korea paused or ended Watson partnerships.

The financial toll was staggering. IBM invested **over \$4 billion** in Watson Health. In January 2022, IBM sold the Watson Health division at a fraction of the original investment -- widely reported as a multi-billion dollar write-down. The root cause, in one sentence: the training data was insufficient in volume, biased toward a single institution's practices, and not representative of the diverse global patient populations Watson was deployed to serve.

The Before and After

WHAT IBM EXPECTED:		WHAT ACTUALLY HAPPENED:	
Training Data:	MSK expert-curated cases + real outcomes	Training Data:	Synthetic/hypothetical cases from one hospital
Coverage:	Global patient diversity	Coverage:	Western patients only, one institution's practices
Result:	Accurate treatment recommendations worldwide	Result:	Unsafe recommendations, dissolved partnerships
Investment:	\$4+ billion	Outcome:	Division sold at massive write-down in 2022

Why This Matters for You

IBM had some of the best engineers on earth. They had billions of dollars. They had partnerships with elite hospitals. None of it mattered because the data was fundamentally flawed. **No algorithm -- no matter how sophisticated -- can compensate for bad data.** This is the principle you will learn to apply in this session: before you ever write a line of analysis, you must ask "Can I trust this data?"

Think About:

- If Watson had been trained on real outcome data from diverse hospitals worldwide, how might the results have been different?
- What responsibility does a data professional have when they know the training data is limited or biased?
- Can you think of other situations where "more data" from a "wider range of sources" would change the quality of a decision?

Key Concepts to Preview

Read through these previews carefully. You do not need to memorise every detail -- the goal is to arrive in class with a general sense of what each concept is, so the discussions and demonstrations feel familiar rather than foreign.

What Is "Garbage In, Garbage Out"?

GIGO is the oldest principle in computing, coined by George Fuechsel, an IBM programmer, in the early 1960s. The idea is simple: if you feed a computer incorrect input data, you will get incorrect output. The computer does not "know" the data is wrong -- it faithfully processes whatever you give it.

In the context of machine learning, GIGO is actually **worse** than the original concept. Traditional programs produce garbage output one-to-one -- one bad input, one bad output. ML models **amplify** garbage: one biased training set produces millions of biased predictions, each with a confidence score that makes the garbage look trustworthy.

The cooking analogy: Imagine you are a world-class chef with decades of experience. Now imagine someone hands you rotten ingredients -- spoiled meat, mouldy vegetables, expired spices. Can your skill save the dish? No. No recipe, no technique, no celebrity reputation can turn rotten ingredients into a safe meal. The dish is ruined before you start cooking. In data science, the analysis is ruined before you write a single line of code if the data is bad.

The GIGO Spectrum -- not all bad data is equally bad:

Severity Level	Type of Problem	Example	Detection
Level 1	Typos and formatting	"Banglore" instead of "Bangalore"	<code>.str.lower()</code> , <code>.value_counts()</code>
Level 2	Missing values	Empty cells where a diagnosis should be	<code>.isnull()</code> , <code>.isnull().sum()</code>
Level 3	Systematic errors	A sensor that consistently reads 2 degrees too high	Requires domain knowledge
Level 4	Fabricated data	IBM Watson trained on hypothetical cases as if they were real	Requires data governance

You already have the Pandas tools for Levels 1 and 2 from Sessions 4.1 and 4.2. This session teaches you the frameworks for Levels 3 and 4 -- problems that code alone cannot solve without human judgment.

Think About:

- Have you ever made a decision based on misleading data -- online reviews, restaurant ratings, college rankings -- and later found out the data was wrong?
- Why might "more sophisticated algorithms" actually make the GIGO problem worse rather than better?

The Six Data Quality Dimensions

Just as a doctor runs a complete health checkup -- checking blood pressure, heart rate, cholesterol, blood sugar, oxygen levels, and BMI -- we need a **complete health checkup for our data**. Checking only for missing values does not mean your data is clean.

#	Dimension	Core Question	Simple Example
1	Accuracy	Does the data match reality?	A patient's age is recorded as 25, but they are actually 52
2	Completeness	Is anything missing?	15 out of 100 patients have no phone number on file
3	Consistency	Do related values agree?	"Mumbai", "mumbai", and "MUMBAI" treated as three different cities
4	Timeliness	Is the data current enough?	Using 2019 salary data to make 2025 hiring offers
5	Validity	Does the data follow the rules?	A patient's age recorded as -5 (age cannot be negative)
6	Uniqueness	Are there duplicates?	The same patient "Priya Sharma" entered twice in the system

A dataset can score perfectly on one dimension and fail catastrophically on another. IBM Watson's training data was arguably "complete" (no missing fields) but failed on "accuracy" (synthetic cases treated as real outcomes). You need to check all six, every time.

An important distinction -- Validity vs Accuracy:

These two dimensions are often confused. Consider a patient's age recorded as 25 in the system:

- **Is it valid?** Yes -- 25 is a perfectly reasonable age that passes all rules (non-negative, under 120).
- **Is it accurate?** Unknown -- the patient might actually be 52. The value 25 "looks right" but could be factually wrong.

Validity says "this COULD be right." Accuracy says "this IS right." Validity is necessary but not sufficient for data quality. You will explore this distinction in detail during the session.

Think About:

- Which dimension do you think is hardest to detect with code alone? (Hint: think about the validity-vs-accuracy distinction above.)
- Can you think of a real-world situation where data was "valid" (followed the rules) but "inaccurate" (did not match reality)?
- If you ran `df.groupby('city')['revenue'].sum()` on a dataset where "Mumbai" and "mumbai" were treated as different cities, what would happen to your results?

Data Governance: Who Is Responsible?

Data governance is the set of policies, processes, roles, and standards that define how data is collected, stored, maintained, and used within an organization.

The shared kitchen analogy: Think of data governance like managing a shared kitchen in a large office building:

- Someone **owns** the kitchen -- the facilities manager sets the rules and budget. This is the **data owner**.
- Someone **keeps it clean daily** -- the cleaning staff wipes counters and checks the fridge. This is the **data steward**.
- There is a **log of what supplies were ordered** -- where the coffee beans came from, which vendor supplies the milk. This is **data lineage**.
- You need **permission to rearrange the appliances** -- you cannot move the refrigerator without approval. This is **change management**.
- **Not everyone has the key** to the locked storage room. This is **access control**.

Every data quality disaster covered in this session -- JPMorgan, the Post Office, gene name corruption -- had governance gaps. Consider what proper governance would have prevented:

- **JPMorgan (\$6.2B loss):** If a data steward had run weekly validation checks on the risk model's output, they would have noticed the risk numbers dropped by half overnight after a formula change. A change management process would have required approval before modifying the formula.
- **UK Post Office (700+ wrongful convictions):** If independent stewards had cross-checked the system's records against physical cash counts, the phantom transactions would have been detected years earlier.
- **Gene name corruption:** If data lineage had tracked file format conversions, researchers would have known their data had passed through Excel and could have checked for corruption.

Governance is not bureaucracy; it is the difference between catching a formula error in a weekly review and losing \$6.2 billion.

Why Documentation Matters

A **data dictionary** is like a nutrition label on packaged food. Without the label, you have no idea what is inside the package -- it could contain allergens, it could be expired, it could be unsafe. You would never eat unlabeled food from a stranger's kitchen. Similarly, you should never analyse an unlabeled dataset.

A data dictionary describes every column: its name, data type, a plain-English explanation of what it represents, the expected range of values, and any constraints or business rules. For example:

Component	Example (for an <code>age</code> column)
Field Name	<code>age</code>
Data Type	<code>float64</code>
Description	Patient's age in years at time of hospital admission
Expected Values	0 to 120 (integers only)
Constraints	Non-null, must be non-negative, must not exceed 120

The flag_42 test: If someone shows you a column called `flag_42` with values 0, 1, 2, 3 -- can you tell what it means? It could be a customer satisfaction score, an error severity level, a priority ranking, or

anything else. Without a data dictionary, you cannot tell. Wrong interpretation leads to wrong analysis -- that is GIGO with extra steps.

In class, you will build both an **automated** data dictionary (what Pandas can generate programmatically using methods like `.dtypes`, `.isnull().sum()`, `.unique()`) and an **enriched** dictionary (what only a human with domain knowledge can provide -- descriptions, business rules, constraints).

Data Quality Is the Law

Data quality is not just a technical best practice -- for many types of data, it is a **legal requirement**. Three major regulations you should be aware of:

Regulation	Jurisdiction	Focus	Key Requirement
GDPR	European Union	Personal data	Data must be accurate and kept up to date
HIPAA	United States	Health data	Confidentiality, integrity, and availability
SOX	United States	Financial data	Accuracy and completeness with audit trails

The penalties for non-compliance are substantial:

- **GDPR:** Up to 4% of global annual revenue or EUR 20 million, whichever is higher
- **HIPAA:** Up to \$1.5 million per year per violation category
- **SOX:** Up to \$5 million in fines and up to 20 years in prison for willful fraud

The building code analogy: Data regulations are like building codes for construction. You CAN build a house without following codes -- it might even look impressive from the outside. But when the earthquake comes (an audit, a lawsuit, a data breach), the house collapses because it was never built to withstand real-world stress. Building codes exist because people were hurt in poorly constructed buildings. Data regulations exist for exactly the same reason: people lost money (Enron, WorldCom), lost freedom (UK Post Office), and lost health (IBM Watson) due to poor data quality. Building codes are not bureaucracy -- they are safety. Data regulations are not paperwork -- they are protection.

As a data professional, you do not need to be a lawyer. But you do need to know that data quality has legal consequences -- and you need to know which regulations apply to the data you are working with.

Warm-up Exercises

Try these short exercises before class. They use Pandas skills you already have from Sessions 4.1 and 4.2 -- the same skills you will apply to data quality checks in the session.

Exercise 1: Checking for Missing Values

Create a small DataFrame and check for missing values -- this is the foundation of the **completeness** dimension.

```
import pandas as pd
```



```
students = pd.DataFrame({
    'name': ['Alice', 'Bob', 'Charlie', 'Diana', 'Eve'],
    'score': [85, None, 92, None, 78],
    'city': ['Mumbai', 'Delhi', None, 'Chennai', 'Pune']
})

# Count missing values per column
print(students.isnull().sum())

# What percentage of each column is missing?
print(students.isnull().mean() * 100)
```

Question: Which column has the highest percentage of missing values? Both `score` and `city` have missing values -- are they equally missing, or does one have a higher percentage?

Why this matters: In class, you will use these exact methods to measure the **completeness** dimension of a healthcare dataset. A column with 12% missing phone numbers tells a different story than a column with 8% missing diagnoses -- the impact on your analysis depends on which data is absent.

Exercise 2: Finding Duplicates

Use `.duplicated()` to identify duplicate rows -- this is the foundation of the **uniqueness** dimension.

```
import pandas as pd

orders = pd.DataFrame({
    'order_id': ['A01', 'A02', 'A03', 'A01', 'A04'],
    'product': ['Mouse', 'Keyboard', 'Monitor', 'Mouse', 'Webcam'],
    'amount': [899, 2499, 8999, 899, 1499]
})

# Find duplicate rows
print(orders.duplicated())

# Show the duplicate entries
print(orders[orders.duplicated(keep=False)])
```

Question: What is the difference between `duplicated()` and `duplicated(keep=False)`? Try both and compare the output. The default marks only the second copy as a duplicate; `keep=False` marks ALL copies -- which is what you need when investigating, because you want to see both records side by side to decide which one to keep.

Why this matters: In class, you will use `.duplicated()` to measure the **uniqueness** dimension. Duplicate patient records can inflate counts ("We have 1,000 patients" when you really have 950) and skew averages.

Exercise 3: Exploring Unique Values

Use `.value_counts()` and `.str.lower()` to check for consistency issues -- this is the foundation of the **consistency** dimension.


```
import pandas as pd

employees = pd.Series(['Mumbai', 'delhi', 'Mumbai', 'DELHI', 'Pune', 'mumbai'])

# How many unique values before cleaning?
print(f"Unique values (raw): {employees.nunique()}")
print(employees.value_counts())

# How many after normalising?
cleaned = employees.str.lower().str.strip()
print(f"\nUnique values (cleaned): {cleaned.nunique()}")
print(cleaned.value_counts())
```

Question: How many "fake" categories does inconsistent capitalisation create? After normalising, how many true unique cities remain?

Why this matters: In class, you will use `.value_counts()` to measure the **consistency** dimension. If a `groupby('city')` on raw data produces 14 city groups instead of 10, every per-city statistic (average revenue, patient count, disease prevalence) is wrong -- not because of bad math, but because the grouping is fragmented by inconsistent formatting.

Setup Requirements

Python Version

Python **3.12.x** (the version you have been using since Week 1).

Required Packages

Pandas is already installed from Session 4.1. No new packages are required.

Verification Code

Run this in a Jupyter Notebook cell or a Python script before class:

```
import pandas as pd
import numpy as np

print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
print("Ready for Session 4.3!")
```

You should see version numbers printed (Pandas 2.x and NumPy 1.x or 2.x) and the message "Ready for Session 4.3!" with no errors.

If you see an `ImportError` or `ModuleNotFoundError`, run `pip install pandas` in your terminal and try again. If you are using Anaconda, use `conda install pandas` instead.

Key Vocabulary

These terms will appear throughout the session. Familiarise yourself with them so they are not completely new when you hear them in class.

Term	Definition
GIGO	"Garbage In, Garbage Out" -- if you feed bad data into a system, you get bad results out
Data Quality	The degree to which data is fit for its intended use across multiple dimensions
Accuracy	Data values correctly represent the real-world entity or event they describe
Completeness	All required data fields are present and populated for all records
Consistency	The same entity is described the same way across all records and systems
Timeliness	Data is current enough for its intended use at the time of analysis
Validity	Data values conform to defined rules, constraints, and formats for their field
Uniqueness	Each record represents a distinct real-world entity -- no duplicates
Data Governance	The policies, processes, roles, and standards for managing data within an organization
Data Stewardship	The hands-on, day-to-day role of monitoring and maintaining data quality
Data Lineage	A record of where data originated and every transformation it has undergone
Data Dictionary	A structured document describing every field in a dataset (type, description, constraints)
GDPR	General Data Protection Regulation -- EU law requiring data accuracy and user consent
HIPAA	Health Insurance Portability and Accountability Act -- US law protecting health data
SOX	Sarbanes-Oxley Act -- US law requiring accuracy and audit trails for financial data
NaN	"Not a Number" -- Pandas representation of missing or unavailable data
<code>.isnull()</code>	Pandas method that returns True for missing values (NaN/None) in a DataFrame or Series
<code>.duplicated()</code>	Pandas method that returns True for duplicate rows in a DataFrame

Things to Ponder

Come to class having thought about these questions. You do not need written answers, but engaging with them beforehand will help you follow the session more actively.

About trust: When you receive a dataset from a colleague or download one from the internet, do you instinctively trust it -- or do you check it first? What would it take for you to trust a dataset enough to make a business decision based on it?

About dimensions: Think of a dataset you have worked with (from an assignment or personal project). Which of the six data quality dimensions would be hardest to verify for that dataset? Why?

About consequences: If a hospital's patient database contained 10% duplicate records, how would that affect decisions about staffing, resource allocation, and treatment capacity? What if the duplicates were not evenly distributed -- what if one department had 30% duplicates?

About responsibility: In the IBM Watson case, who bears responsibility for the data quality failure -- the engineers who built the system, the doctors who curated the training data, the executives who approved the deployment, or the hospitals that adopted it without independent validation? Is there a single "right" answer?

About tools: Scientists had to rename human genes because Excel kept corrupting them (converting gene name "SEPT2" to "September 2"). Pandas does not auto-format your data -- it preserves strings exactly as you give them. Can you think of other situations where a tool's "helpful" automatic behaviour caused more harm than good?

Pre-Class Checklist

Before you walk into class, make sure you can check off every item:

- ☐ I have Python 3.12.x installed with Jupyter Notebook (or JupyterLab)
- ☐ I can `import pandas as pd` and `import numpy as np` without errors
- ☐ I have read the IBM Watson for Oncology business story and understand why data quality matters
- ☐ I know what GIGO means and can explain it in one sentence
- ☐ I can name at least four of the six data quality dimensions
- ☐ I understand the difference between "completeness" (is data missing?) and "validity" (does data follow rules?)
- ☐ I remember how to use `.isnull().sum()` to count missing values (Session 4.2)
- ☐ I remember how to use `.value_counts()` to explore unique values (Session 4.2)
- ☐ I remember how to use `.str.lower()` and `.str.strip()` for string cleaning (Session 4.2)
- ☐ I have reviewed the vocabulary table above

Looking Ahead

After this session, you will be able to:

- Evaluate any dataset across six quality dimensions before starting analysis
- Detect missing values, duplicates, inconsistencies, invalid entries, and stale records using Pandas

- Explain at least five real-world data quality disasters and their specific consequences
- Build a data dictionary that documents a dataset's columns, types, constraints, and descriptions
- Produce a quantitative quality scorecard summarising a dataset's fitness for analysis
- Understand the basics of data governance and why organizations need formal frameworks for managing data quality

The mini-project: During the session, you will apply all of these skills in a hands-on Data Quality Audit Report. You will receive a messy healthcare dataset (`patient_records`) with intentional quality issues spanning all six dimensions, run Pandas-based checks for each dimension, and assemble the results into a structured scorecard summarising the dataset's fitness for analysis.

The 80/20 reality: Industry surveys consistently show that data scientists spend approximately **80% of their time on data preparation** (collection, quality auditing, cleaning, standardisation) and only **20% on model development** (training, tuning, evaluating). As Andrew Ng, Stanford professor and co-founder of Coursera, has said: "The most important thing in machine learning is the data." The skills you are building in this session -- quality assessment, governance, documentation -- are not prerequisites you rush through to get to the "real stuff." These ARE the real stuff.

This session completes the **data foundation** you have been building across Weeks 3 and 4. In **Session 5.1**, you will learn **data visualization** -- creating charts, plots, and visual summaries that communicate your findings. But remember what you will learn in this session: visualization amplifies whatever is in your data. If the data is clean, your charts reveal real insights. If the data is dirty, your charts create beautiful, convincing lies. The quality skills from this session are the filter that separates truthful visualizations from dangerous ones.

See you in class!

Vishlesan i-Hub IIT Patna x Masai School